Learning Objectives

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Learning Objective 1: Identify the factors released by endothelial cells and its effects on vascular function

Learning Objective 2: Describe the main role and differences between endothelial and vascular smooth muscle cells

Learning Objective 3: Summarize the main signaling pathway involved in the release of nitric oxide from endothelial cells.

Learning Objective 4: Describe endothelial dysfunction and its associated complications

Vascular Structure

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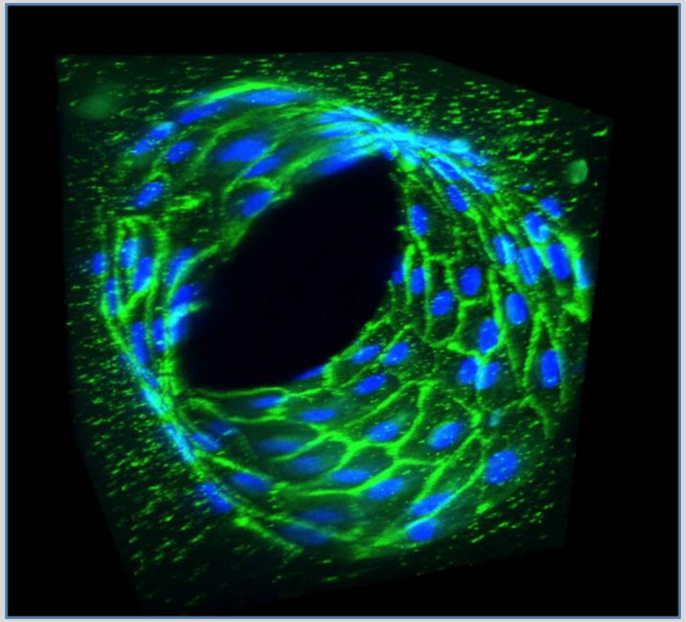
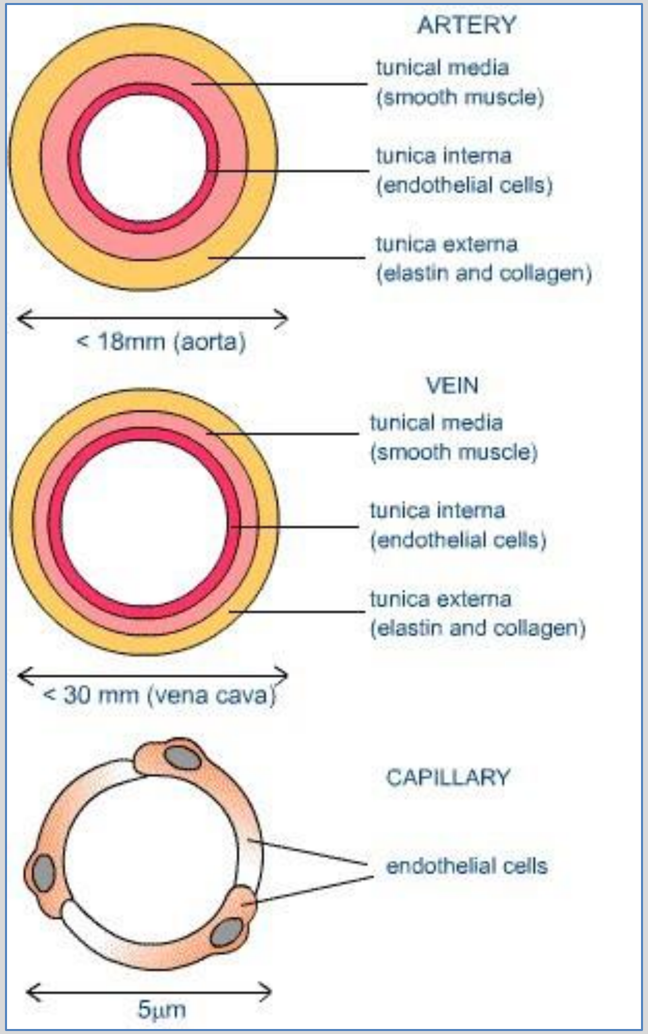
- Vessel layers
  - **Tunica interna (endothelium)**
  - Tunica media (smooth muscle)
  - Tunica externa (elastin/collagen)
- Arteries have thicker smooth muscle layers than veins (<u>think about their function in relation to the structure</u>)
- Endothelium is **monolayer** (smooth muscle is multilayer)
  - Green=endothelial cells
  - Blue=nuclei
- **Endothelium is a semipermeable membrane** to allow for exchange of nutrients and oxygen especially through the capillaries





Where do you find endothelial cells?

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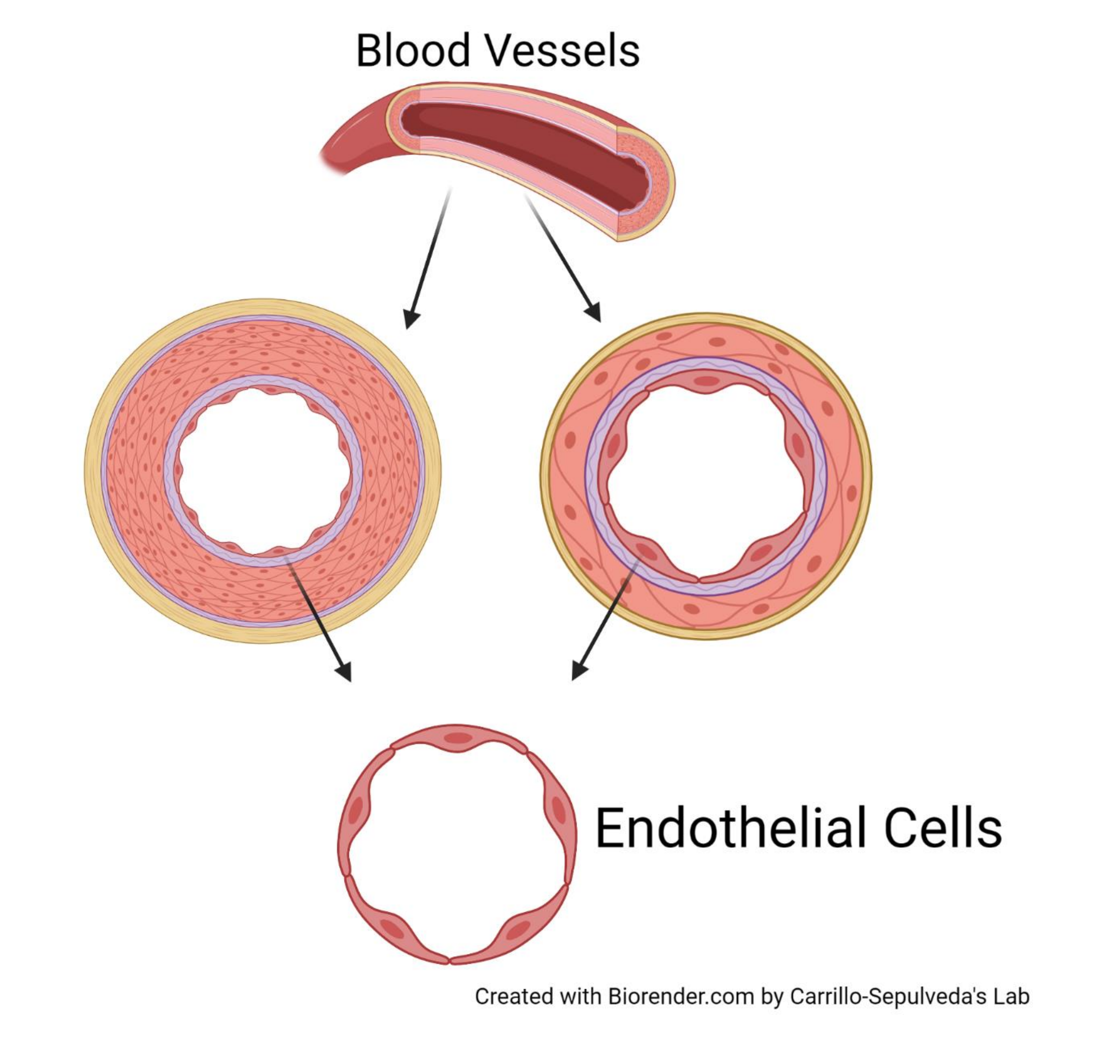
- Endothelial cells are found in **all** blood vessels (arteries and veins, large and small)
- Veins: thinner, weaker
- Arteries: thicker, stronger
- **Compose the inner layer of vessels (endothelium)**



Endothelial Cells

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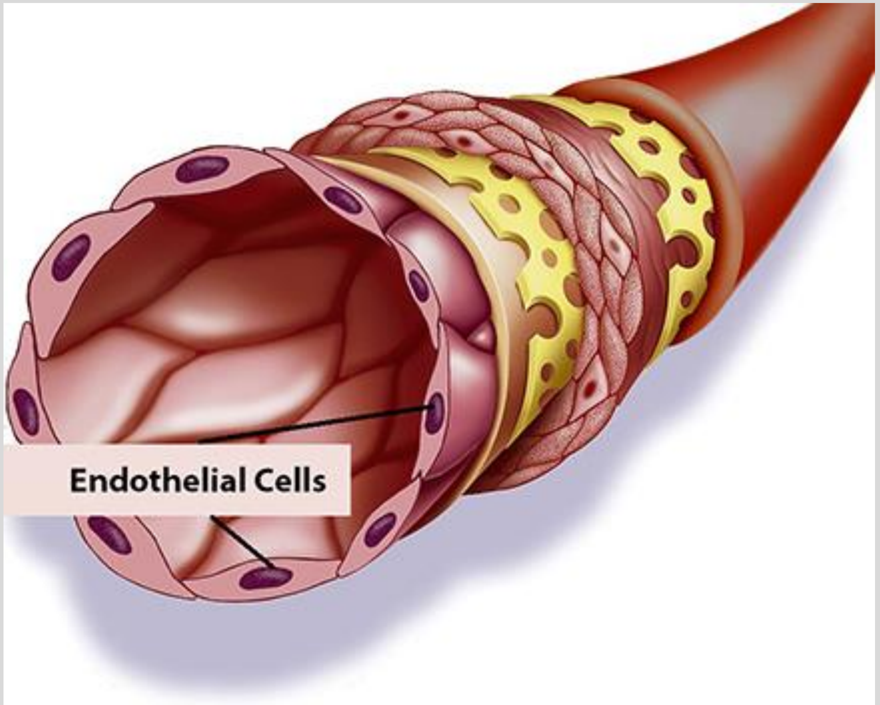
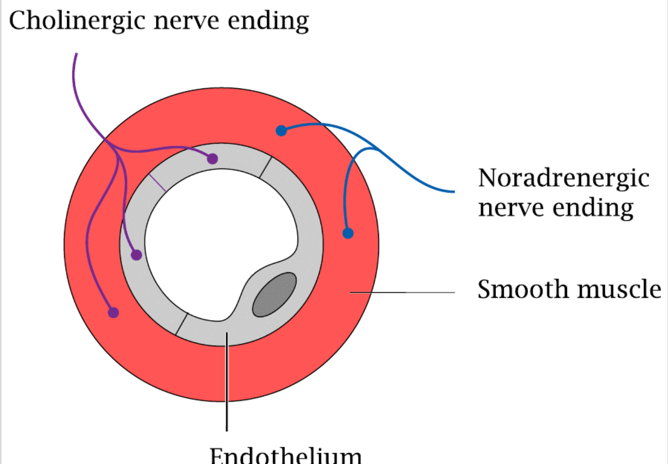
- Primary function of endothelial cells is a **physical barrier**
  - Additional research has discovered a much larger role of endothelial cells in human physiology
- **Endothelium is a monolayer of endothelial cells that lines the blood vessels**
- There are approximately 60 trillion endothelial cells in the human body.
- The **endothelium is innervated by cholinergic nerves and respond to acetylcholine (ACh)**
  - The vascular smooth muscle cell (VSMC) layer is primarily innervated by noradrenergic nerves and responds to epinephrine





Vasoregulation by Endothelium

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- **Balance between vasodilators and vasoconstrictors maintains physiologic conditions**
- Any **disruption** to the balance will cause conditions with either <u>too much vasodilation or too much vasoconstriction</u>

Nitric Oxide
Prostacyclin

Vasodilation

Reactive Oxygen Species
Endothelin-1
Thromboxane (TXA_2)

Vasoconstriction

Endothelial Dysfunction

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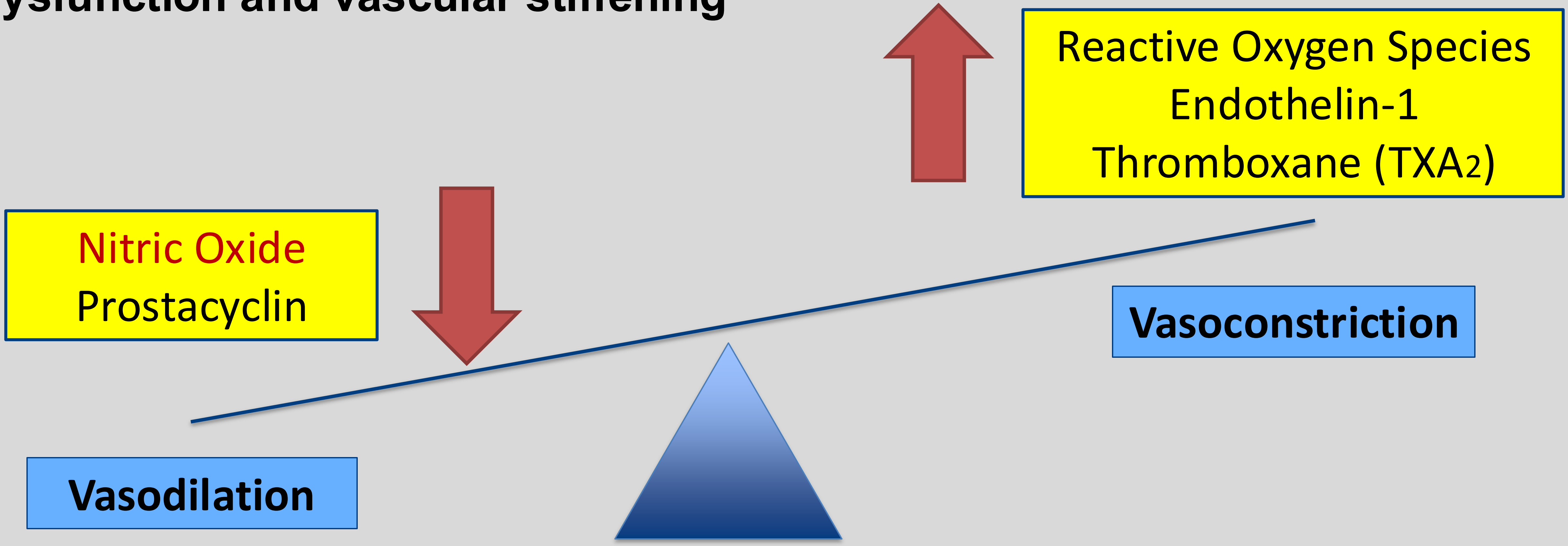
- Endothelin-1 and TXA_2 induce migration and proliferation of VSMC
- Nitric Oxide has anti-proliferative properties
- **Endothelial Dysfunction leads to vascular dysfunction and vascular stiffening**



Vasodilation by Endothelium-derived relaxation factor (EDRF)

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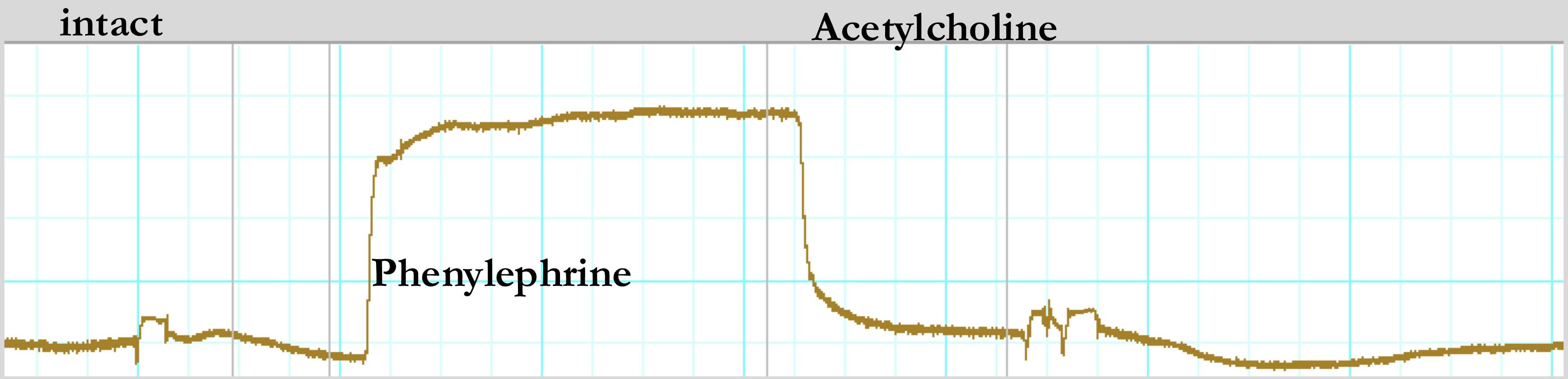
♥ **Nitric Oxide (NO)** is an EDRF that causes **vasodilation**

♥ Acetylcholine (ACh) induces the release of NO leading to relaxation

♥ When the vessel is damaged, the integrity of the endothelial layer is compromised.

What will happen to the relaxation curve in the damaged vessel?



Vasodilation by Endothelium-derived relaxation factor (EDRF)

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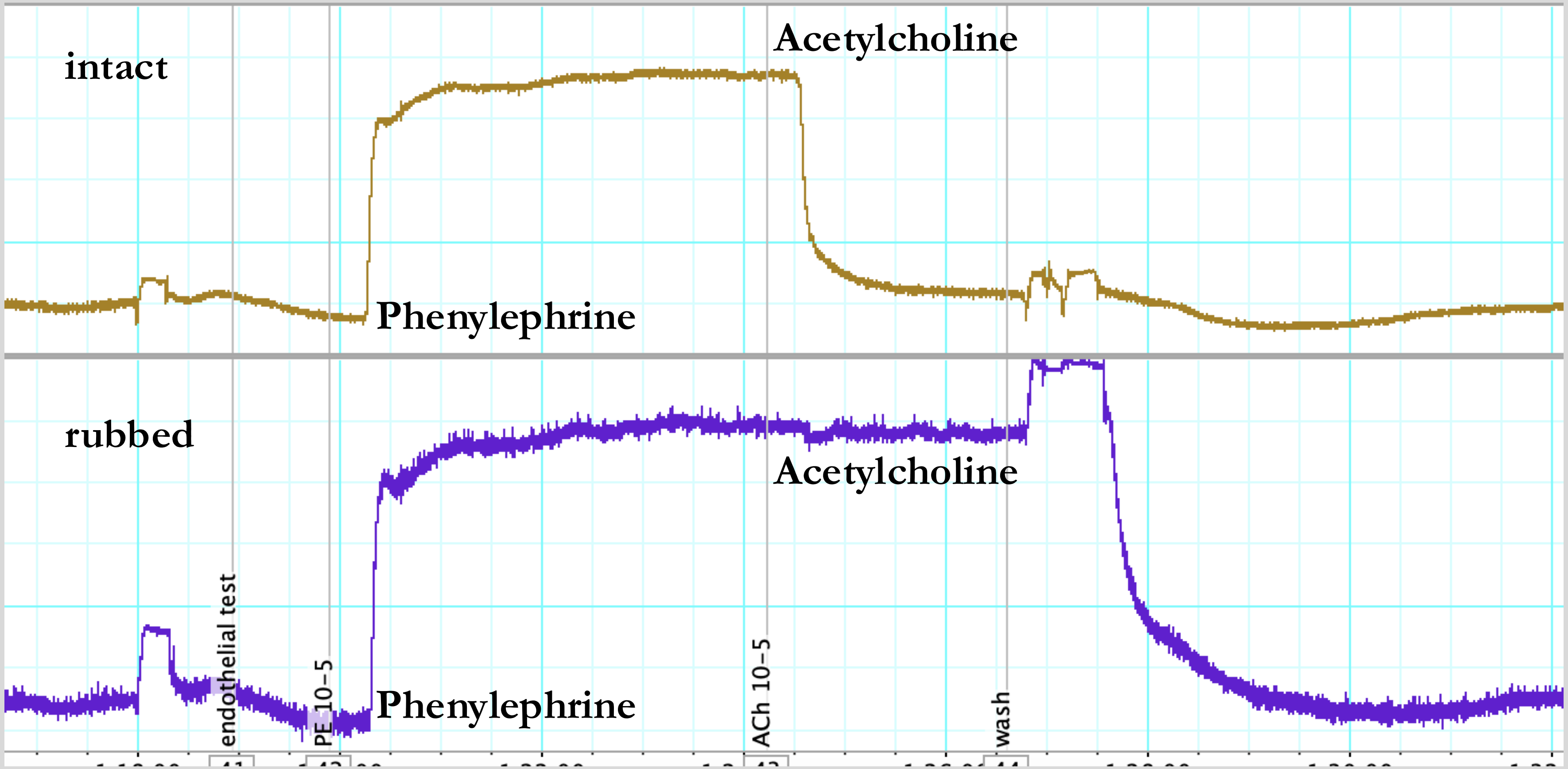
- Nitric Oxide (NO) is an EDRF that causes **vasodilation**
- ACh induces the release of NO leading to relaxation
- When the vessel is rubbed, the endothelial layer is damaged/removed

What will happen to the relaxation curve in the damaged vessel?

- The vessel will not relax because NO is not being released from endothelial cells



Nitric Oxide Signaling

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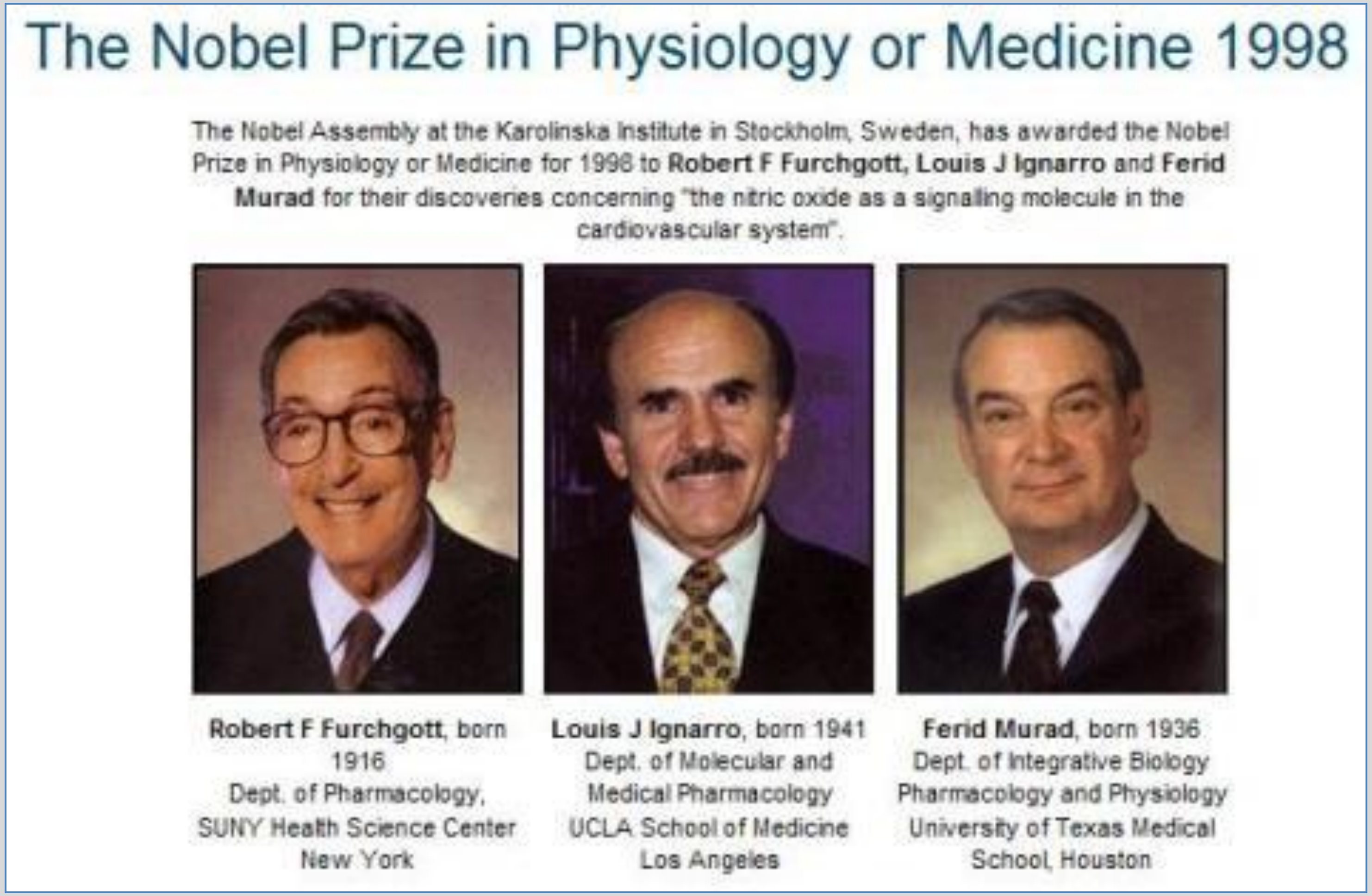

For their discoveries concerning nitric oxide as a signaling molecule in the cardiovascular system

- NO is an important signaling molecule in vascular function
- **NO comes from endothelial cells**
- This discovery later went on to assist in the development of multiple drugs such as Viagra

Nitric Oxide Signaling Cascade

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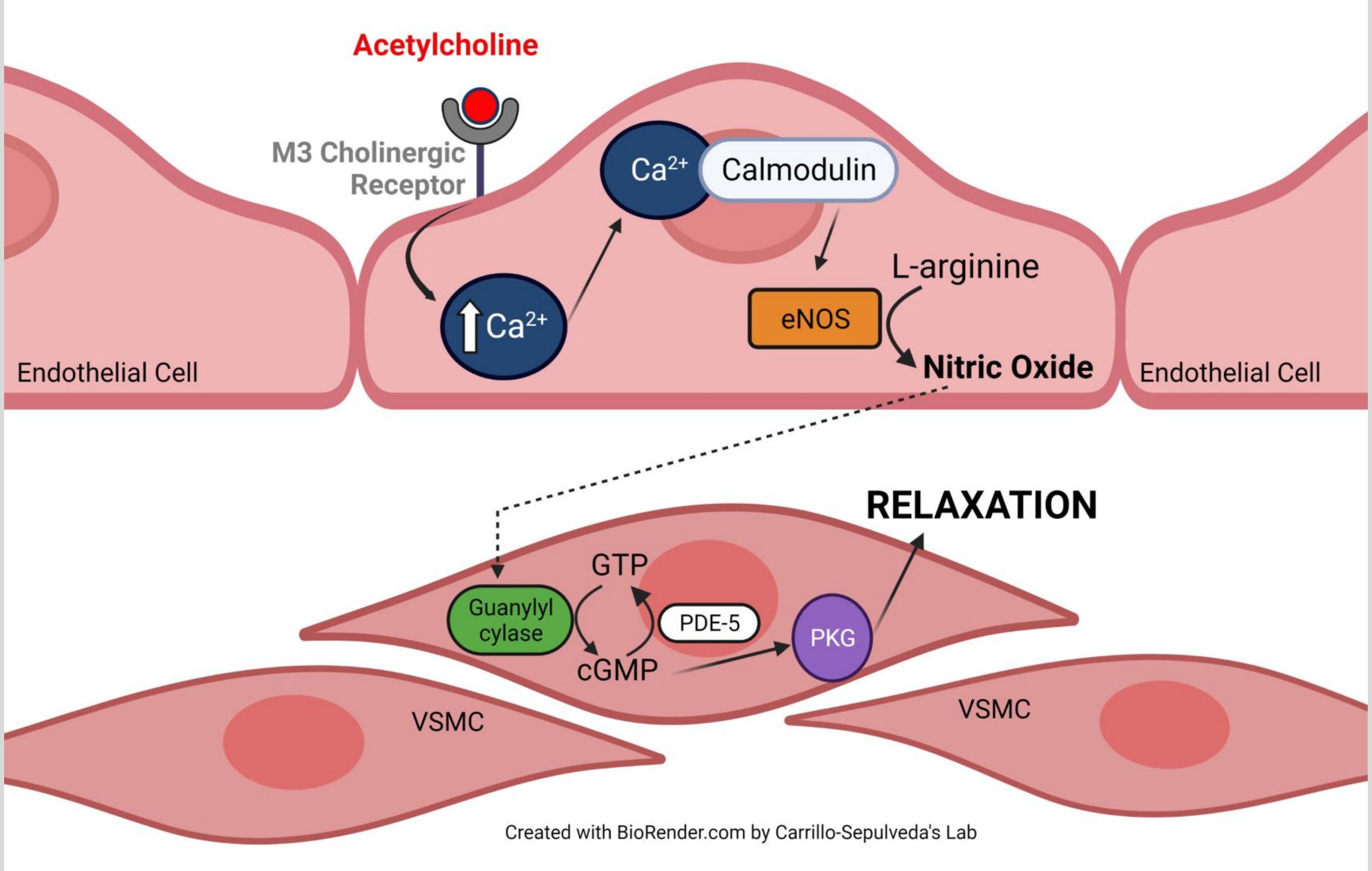

1. **ACh binds the M3 receptor** on endothelial cells and releases Ca^{2+} into the cell
2. Ca^{2+} binds to calmodulin, activating endothelial nitric oxide synthase (eNOS)
3. **eNOS converts L-arginine into NO**, which diffuses into VSMCs
4. **NO activates GC to convert GTP into cGMP**
5. **cGMP activates PKG to cause relaxation**

Nitric Oxide Signaling Cascade

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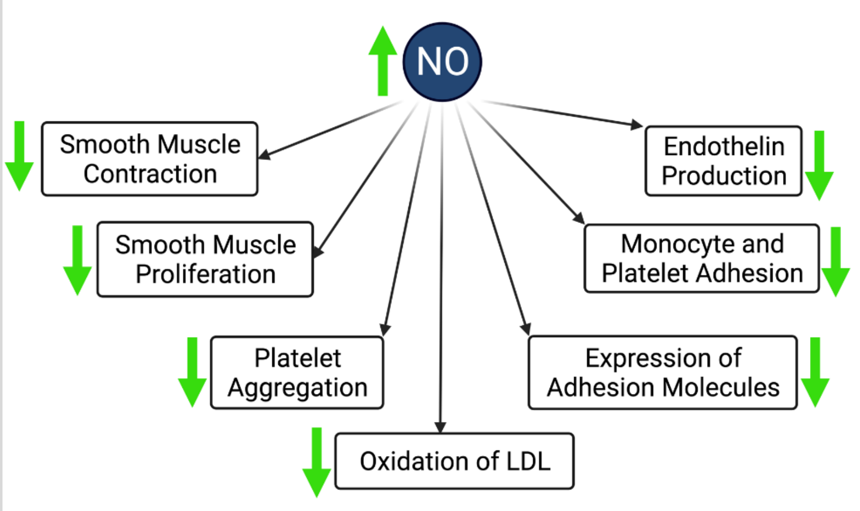
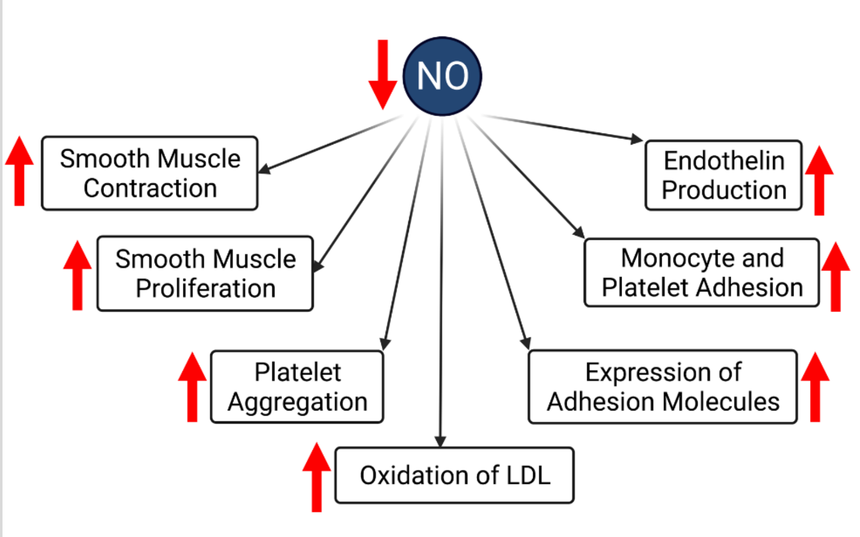
Normal Endothelial Cells



- NO production causes more than just vasodilation

Dysfunctional Endothelial Cells



- **Low NO production** causes vasoconstriction, **smooth muscle proliferation**, etc.
- Seen in many disease states

Severe Allergic Reactions

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- Allergen exposure leads to increased levels of bradykinin (inflammation) and histamine (from mast cells)
- **Bradykinin and histamine cause vasodilation by increasing NO production in endothelial cells**
- Treatment with epinephrine reverses the reaction by causing vasoconstriction
- **Epinephrine activates alpha adrenergic receptors in the smooth muscle layer**

Von-Willebrand Disease

- **Von Willebrand Factor (vWF) is a coagulation factor only released by endothelial cells.**
- **Von Willebrand disease is an autosomal dominant genetic disease characterized by decreased vWF.**
- vWF binds to factor VIII and platelets to assist in the formation of a platelet plug

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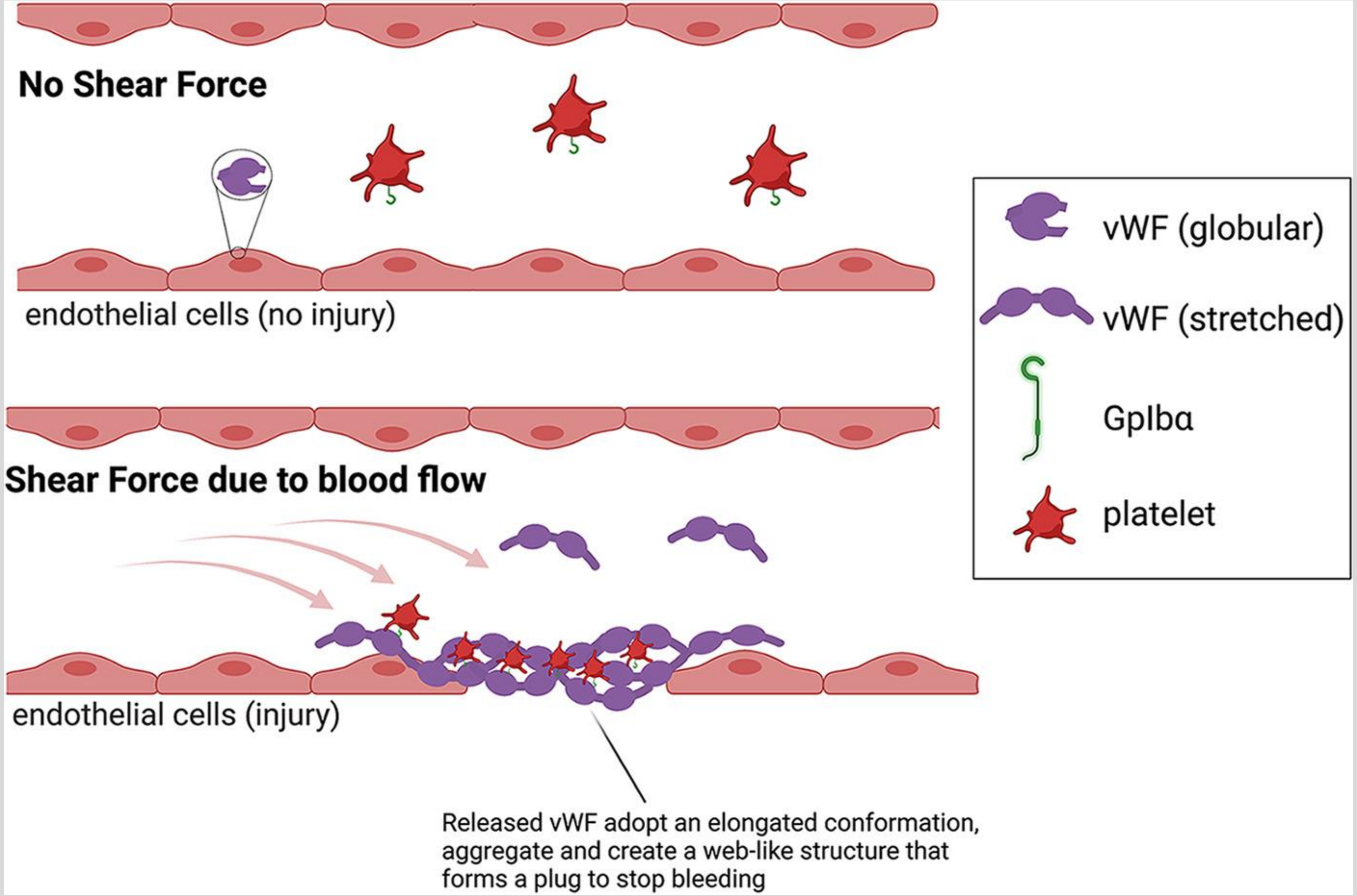

Webster SS, Wong GCL, O'Toole GA. The Power of Touch: Type 4 Pili, the von Willebrand A Domain, and Surface Sensing by Pseudomonas aeruginosa. *J Bacteriol*. 2022;204(6):e0008422. doi:10.1128/jb.00084-22

- Treatment is **Desmopressin**, which releases vWF and Factor VIII from endothelial cells

Hypertension

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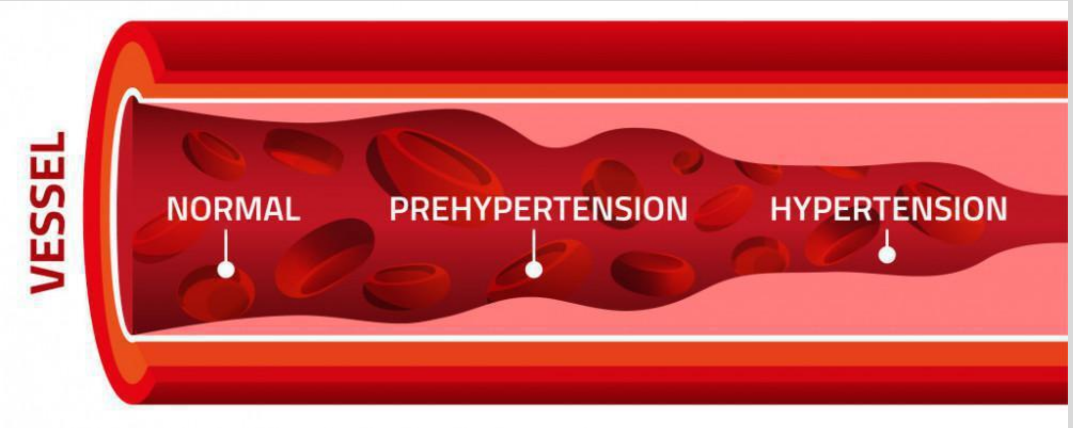

- Endothelial dysfunction causes hypertension (HTN)
- **Decreased levels of NO causes increased vasoconstrictor tone leading to increased BP**
- HTN can also cause additional endothelial damage by shear stress forces

Pulmonary Arterial Hypertension

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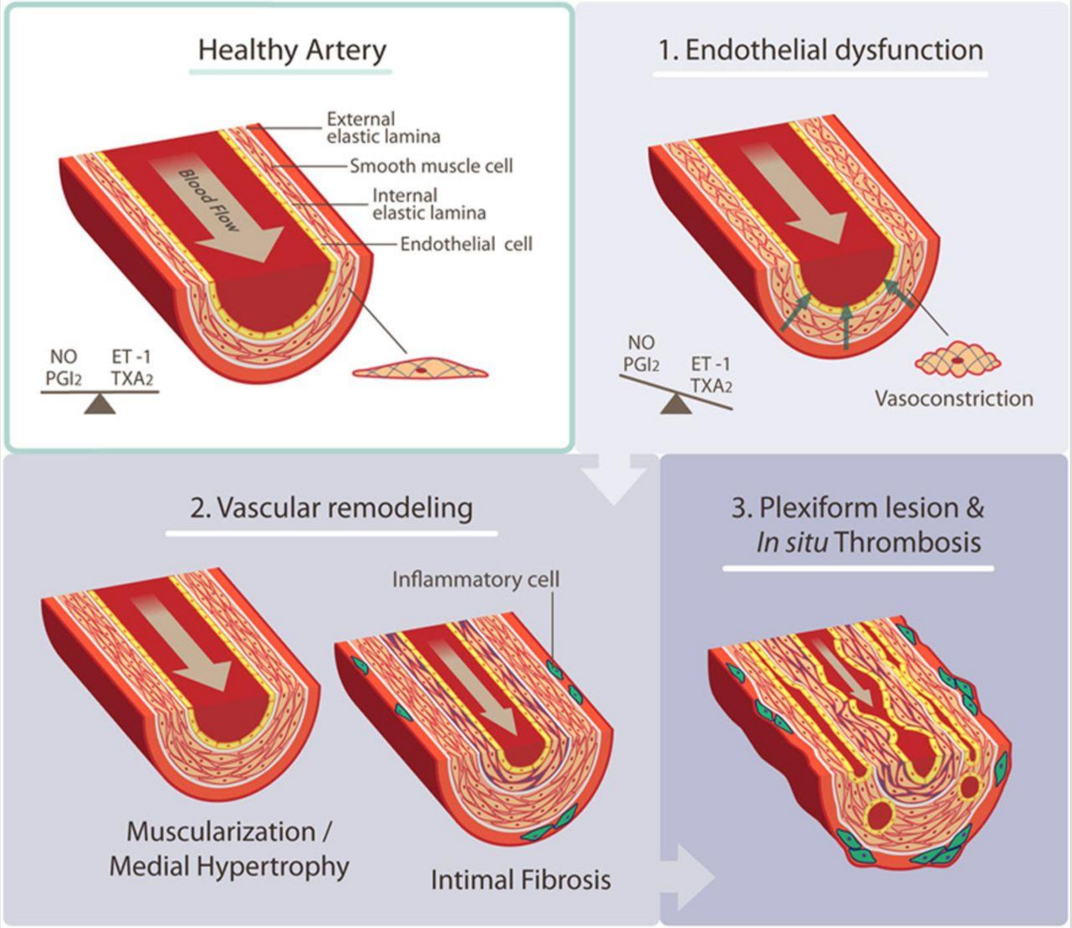
- Elevated mean pulmonary artery pressure (>20 mmHg) at rest
- PAH is associated with endothelial dysfunction in pulmonary vasculature leading to **increased vasoconstrictors (endothelin) and decreased vasodilators (NO and prostacyclins)**
  - Treat with **bosentan** (endothelin-1 receptor antagonists) or **Sildenafil** (Viagra; PDE-5 inhibitor)



Lai YC, Potoka KC, Champion HC, Mora AL, Gladwin MT. Pulmonary arterial hypertension: the clinical syndrome. *Circ Res*. 2014;115(1):115-130. doi:10.1161/CIRCRESAHA.115.301146

Type II Diabetes Mellitus

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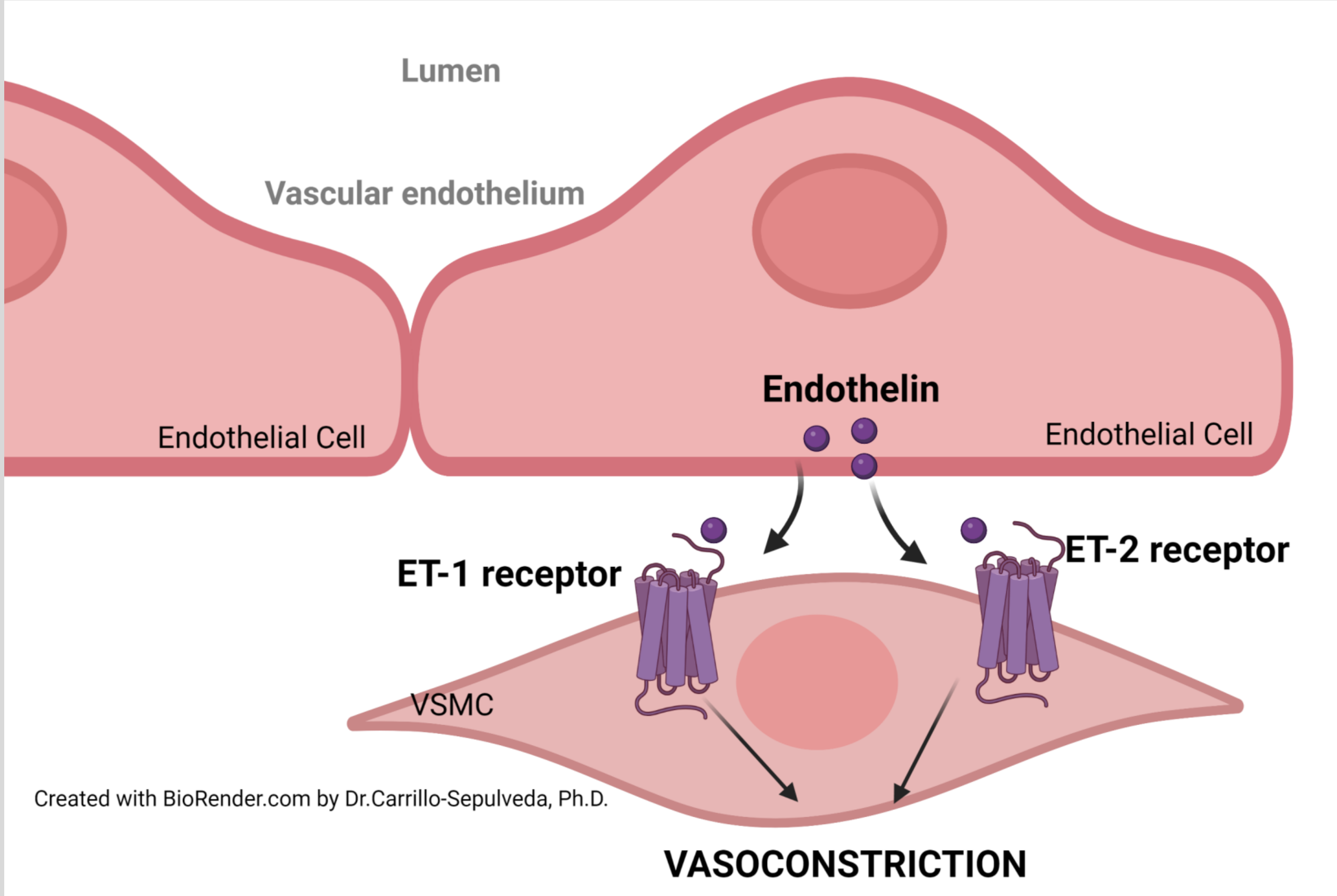

♥ Endothelial Dysfunction
♥ Less NO production
♥ Endothelin-1 production is increased in T2DM
♥ **Endothelin-1 activates ET_A and ET_B on VSMC to cause vasoconstriction and smooth muscle proliferation**

Obesity

- Many believed that all endothelial damage came from inside the vessel ("inside-out" theory of damage)
- There is now research supporting an **"outside-in" theory of endothelial damage**
- The fat surrounding the vessels (perivascular adipose tissue) can become dysfunctional and release factors that contribute to endothelial dysfunction and subsequent HTN
- More research is needed to determine the mechanisms involved

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Lean Obese

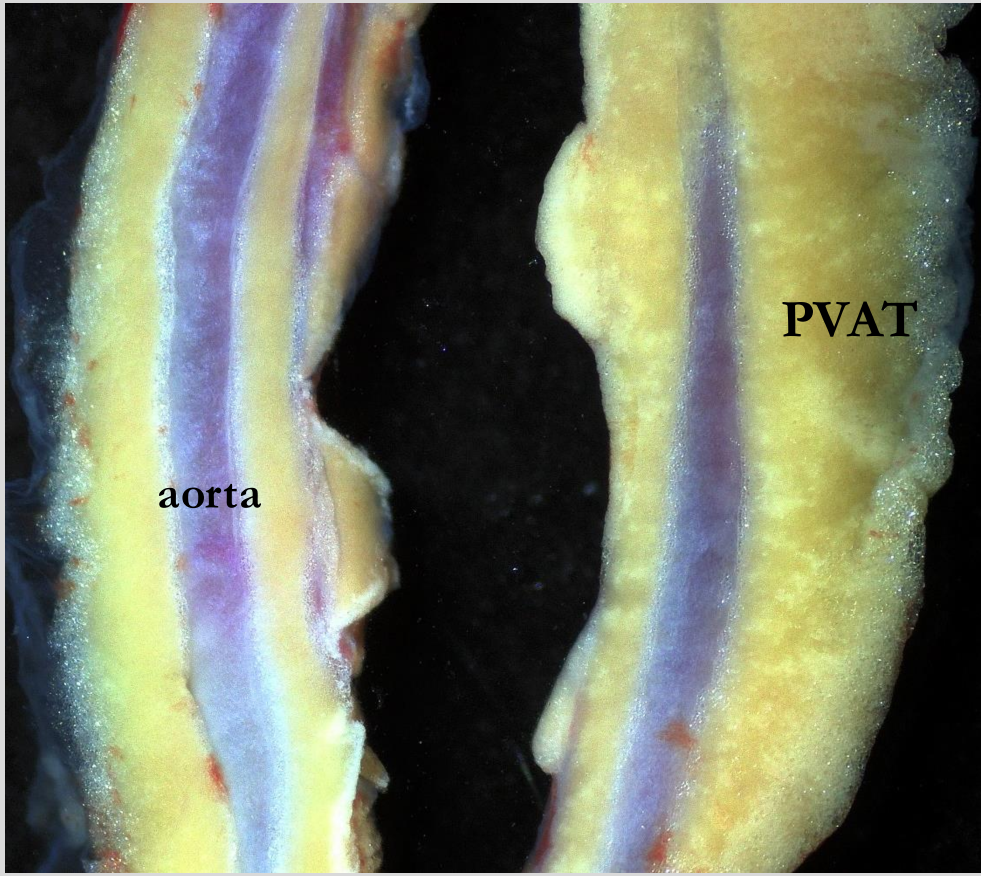

Image from NYITCOM – Sepulveda's Laboratory

Treatment Options

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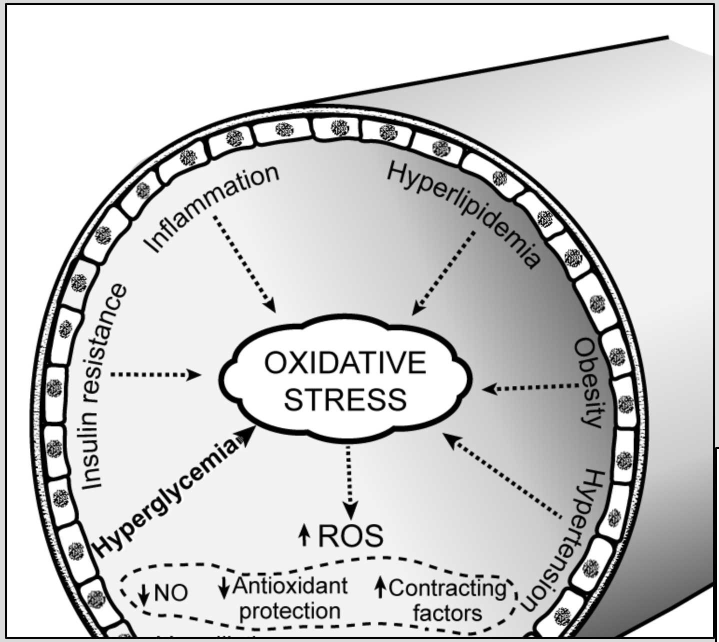
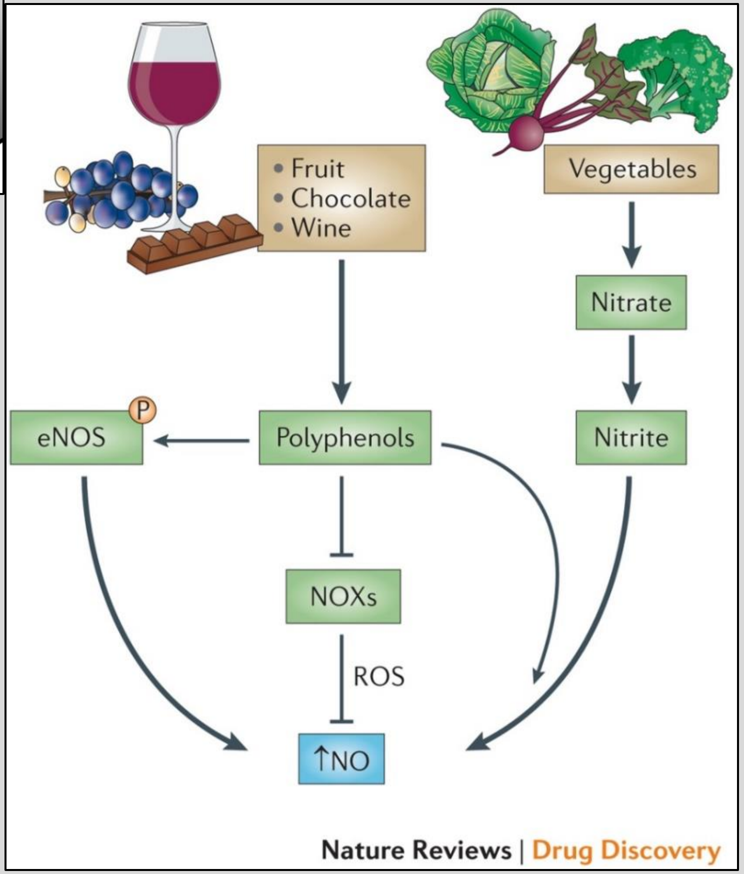


♥ Endothelial dysfunction leads to **increased production of ROS leading to oxidative stress**

♥ Oxidative stress is commonly seen in diabetes, HTN, inflammation and obesity



♥ Oxidative stress results in **increased vasoconstriction (less vasodilation)**

♥ Possible treatment is to consume **anti-oxidant foods and medications to prevent the production of ROS**

♥ These worked well in culture and lab studies but failed during human trials

♥ **Phenols** are thought to block ROS production by **increasing eNOS and blocking the NOX proteins that produce ROS**

Practice Question

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A 51-year-old male patient presents to the emergency department complaining of headache and blurred vision. Physical and lab examination revealed blood pressure of 160/95mmHg, A1C 8%, fasting glucose levels of 240mg/dL. Patient has history of coronary disease. Which of the following most likely explain the cardiovascular complication found in this patient?

a. Increased production of nitric oxide
b. Increased production of prostacyclin
c. Decreased production of reactive oxygen species
d. Decreased production of endothelin-1
e. Increased production of reactive oxygen species

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d. Decreased production of endothelin-1
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Practice Question

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A 63-year-old woman with a history of poorly controlled hypertension and type 2 diabetes is found to have reduced flow-mediated dilation on vascular ultrasound, indicating impaired endothelial function. Which of the following most likely contributes to her elevated blood pressure?

a. Decreased endothelin-1 production
b. Increased nitric oxide release
c. Decreased nitric oxide availability
d. Increased prostacyclin production
e. Decreased sympathetic tone

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c. **Decreased nitric oxide availability**
d. Increased prostacyclin production
e. Decreased sympathetic tone

Summary

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- Endothelium is an endocrine organ, releasing multiple factors to maintain balance between vasodilation and vasoconstriction
- Endothelial cells are mononuclear, grow in a monolayer and are innervated by M3 cholinergic neurons
- **Nitric oxide** as a key signaling molecule in the cardiovascular system causing VSMC vasodilation
- Sildenafil (Viagra) promotes vasodilation by blocking cGMP degradation
- Endothelial dysfunction is a hallmark of many cardiovascular diseases including HTN, Pulmonary HTN, Obesity, T2DM and DVT and mostly recently COVID-19
- Endothelial cells are the exclusive producer of vWF and play an important role in VWD pathogenesis